

Tarea 6.5

$$3) \cos 64 \cos (-44)$$

$$\begin{aligned}\cos 64 \cos (-44) &= \frac{1}{2} [\cos (64 + (-44)) + \cos (64 - (-44))] \\ &= \frac{1}{2} [\cos 24 + \cos 104]\end{aligned}$$

$$\cos 64 \cos (-44) = \frac{1}{2} \cos 24 + \frac{1}{2} \cos 104$$

$$11) \cos 5x - \cos 3x$$

$$\cos 5x - \cos 3x = -2 \sin \frac{5x+3x}{2} \sin \frac{5x-3x}{2}$$

$$\cos 5x - \cos 3x = -2 \sin 4x \sin x$$

$$19) \frac{\sin u + \sin v}{\cos u + \cos v} = \tan \frac{1}{2}(u+v)$$

$$\frac{2 \sin \frac{u+v}{2} \cos \frac{u-v}{2}}{2 \cos \frac{u+v}{2} \cos \frac{u-v}{2}} = \tan \frac{1}{2}(u+v)$$

$$\frac{\sin \frac{u+v}{2}}{\cos \frac{u+v}{2}} = \tan \frac{1}{2}(u+v)$$

$$\tan \frac{1}{2}(u+v) = \tan \frac{1}{2}(u+v)$$

$$25) (\sin ax)(\cos bx)$$

$$\frac{1}{2} [\sin(ax+bx) + \sin(ax-bx)]$$

$$(\sin ax)(\cos bx) = \frac{1}{2} \sin[(a+b)x] + \frac{1}{2} \sin[(a-b)x]$$

$$31) \cos 3x + \cos 5x = \cos x$$

$$2 \cos \frac{3x+5x}{2} \cos \frac{3x-5x}{2} = \cos x$$

$$2 \cos 4x \cos -x = \cos x$$

$$2 \cos 4x \cos x - \cos x = 0$$

$$\cos x (2 \cos(4x) - 1) = 0$$

$$\cos x = 0$$

$$x = \cos^{-1} 0 = \frac{\pi}{2} + 2\pi n$$

$$= \pi + 2\pi n$$

$$2 \cos(4x) - 1 = 0$$

$$\cos 4x = \frac{1}{2}$$

$$4x = \cos^{-1} \frac{1}{2} = \frac{\pi}{3} + 2\pi n$$

$$= \frac{5\pi}{3} + 2\pi n$$

$$\frac{\pi}{2} + \pi n, \frac{\pi}{2} + \frac{\pi}{2} n, \frac{5\pi}{2} + \frac{\pi}{2} n$$